

Sponsorship effect metric: assessing the financial value of sponsoring by comparisons to television advertising

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Abstract Since sponsoring is an increasingly important marketing communication medium, it is important for managers to know whether their sponsorships are giving a good financial return relative to other communication alternatives. Although several “rules of thumb” are commonly used in the industry for converting sponsorship exposure time into television advertising equivalent values, this research is the first to empirically test them by means of two experiments employing realistic stimuli and representative samples. Across 6 stimuli brands, study 1 finds that an average of 127 s of sponsor logo exposure achieves the same effects as a 30 s TV ad, but variance around this figure is high and depends on a variety of factors such as communication goal, sponsor fit, type of exposure, and whether the sponsor is a known brand. Study 2 finds that combining sponsoring with television advertising does increase the effects of sponsoring, although effects vary widely depending on the context, but even in the best case the increase is not supportive of the high levels of advertising suggested by many industry experts. Overall, the findings will allow managers to more confidently assess the financial value of their sponsorship investments.

Keywords Sponsoring · Advertising · Sponsoring conversion rate

Introduction

Determining the value received from marketing investments such as communication campaigns is a growing focus of corporate and academic efforts (Briggs et al. 2005; Mizik and Jacobson 2008). Although sponsoring is an increasingly important marketing communication tool, one of the most frequent criticisms leveled at the industry has been the lack of attention paid to measuring sponsorship effects relative to the investments made (Crompton 2004; Currie 2004). The most common type of sponsoring effects research involves counting seconds of viewer exposure time to the sponsor’s logo shown on the perimeter boards that surround the playing field or covering the team’s uniform during coverage of the sponsored object (i.e. event, organization, or individual). This sponsor logo exposure is then converted via various “black-boxes” into TV advertising equivalencies using widely varying and/or proprietary conversion values ranging from 5% to 100% (Cornwell et al. 2005; Crimmins and Horn 1996; Crompton 2004; Harvey 2001; Meenaghan 2001). Other “rules of thumb” suggests that achieving sponsoring effects requires linking the sponsoring brand more closely to the sponsored object and/or “activating” the sponsorship with advertising (Cornwell et al. 2005, Johar et al. 2006).

Unfortunately these sponsorship “black boxes” and “rules of thumb” are almost totally devoid of empirical verification, making it difficult to provide managers with a solid basis for determining the financial value of sponsoring (Currie 2004; Harvey 2001). The current research takes the first steps towards empirically determining the value of sponsoring versus and together with TV advertising using experiments that employ highly realistic simulated TV coverage of a sponsored sporting event, real 30 s TV spots, market representative samples, and industry relevant meas-

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ures of advertising/sponsoring effects. The paper is organized as follows: Study 1 opens up the “black-box” by empirically determining the conversion rate between sponsoring and TV advertising, which is followed by study 2’s investigation of the common industry suggestion that sponsors need to heavily invest in advertising to activate their sponsorships and create “synergy” effects. Our findings provide empirically grounded estimates to guide managerial decisions and directions for future research.

Study 1: Comparing sponsoring to TV advertising

Simply counting seconds of clear logo exposure during a sponsored event and valuating it based on the costs to reach the same audience via television ads is popular in large part because it is intuitively easy for managers to understand (Cornwell et al. 2005; Crompton 2004; Currie 2004). Another reason that TV advertising provides a reasonable basis for sponsorship valuations is that the communication goals for both are frequently very similar (Harvey 2001). Much of advertising theory, research, and managerial concern has focused on variations of hierarchy of effects models that typically begin with awareness and end with a purchase. While the advertising literature provides little empirical support for such time sequenced hierarchical effects, it does support the use of its components as time independent effects of advertising (Vakratsas and Ambler 1999). Similarly, Marketing Evolution, one of the leading market research firms specializing in measuring advertising effects, typically employ three metrics of success: 1) brand awareness, 2) brand image, and 3) intent to purchase (Briggs et al. 2005). Sponsors frequently have very similar communication goals revolving around cognitive, affective, and behavioral outcomes (Cornwell et al. 2005; Rifon et al. 2004; Thørmøe et al. 2002; Verity 2002).

In general practice, the unverified conversion rates cited previously are typically utilized in a “one size fits all” manner, and hence no attempt is made to adjust the rates to match possible variations in sponsoring effectiveness due to the particular circumstances of the sponsorship context (Harvey 2001). Previous research suggests a number of variables may influence the effectiveness of sponsoring including: 1) the type of communication goal, 2) whether the sponsoring brand is previously known by the audience, 3) the fit of the sponsor with the sponsored object, and 4) the type of exposure the sponsor receives.

Although the communication goals for advertising and sponsoring are frequently similar, from the viewpoint of the audience member, sponsoring is different than advertising in a variety of ways. First, the movement of the TV camera often chops up audience exposure to the sponsor during the event as it follows the action on the field, which contrasts

with advertising that is more concentrated within one or more 30-second spots. Second, advertising usually involves a “message” of some type that will typically require some level of cognition and comprehension to process, and its persuasiveness is typically highest in situations where the audience has little previous knowledge about the advertised brand or product (Cornwell, et al. 2005; Crompton 2004; Vakratsas and Ambler 1999). In contrast, sponsor exposure is typically passive with simple perimeter boards around the playing field, and persuasion is indirect through “mere exposure effect” and image and affect transfer from object to sponsor (Crimmins and Horn 1996; Gwinner and Eaton 1999; Olson and Thørmøe 2003). Third, in contrast to advertising that is likely most persuasive for new brands, previous research has found that the persuasive abilities of sponsoring is likely to be hampered for less known brands because they are less likely to be identified as the sponsor (Johar and Pham 1999; Johar et al. 2006). Although literature has noted how the mediums differ regarding audience experience, cognition requirements, and persuasion processes, no previous research has empirically verified the relative effectiveness of sponsoring versus TV advertising. Thus, research questions remain about how different communication goals and the degree to which the brand is known might create variance in the relative effectiveness of the two mediums.

RQ1: Does the relative effectiveness of sponsoring versus TV advertising vary depending on the communication goal (i.e. brand recognition, attitude change, purchase intention)?

RQ2: Does the relative effectiveness of sponsoring versus TV advertising vary depending on whether the sponsoring brand is well-known or unknown?

The most frequently investigated theoretical concept related to sponsoring is the fit / relatedness / congruence between the sponsor and object, which can be based on the sponsor’s relevance to the object, functional similarities, target audience similarities, and/or image/symbolic similarities (Cornwell et al. 2005). Fit is theorized to influence the frequency, depth, and valence of attributions that consumers give to the sponsor for supporting the object, with high fit leading to more seamless acceptance of the brand’s altruistic motives for sponsoring. Conversely, a relative lack of fit stimulates cognitive evaluation and elaboration that can yield greater resistance to the positive sponsorship message (Rifon et al. 2004). There is some mixed support as to the benefits of higher fit in sponsoring contexts, particularly with regards to sponsor recognition accuracy. Jagre et al. (2001), contend that lower levels of fit are preferable to high fit because less related sponsors would stand out more and hence increase the likelihood of stimuli encoding and recall from memory. The majority of empirical research, however, has found a

positive relationship between sponsor fit and sponsoring effects of all types (Cornwell et al. 2005; Martensen et al. 2007; Simmons and Becker-Olsen 2006; Speed and Thompson 2000), although a research question remains about fit's importance in determining relative effectiveness of the two mediums.

RQ3: Does the degree of fit between sponsor and object influence the relative effectiveness of sponsoring versus TV advertising?

In part to enhance fit perceptions, sponsor industry consultants frequently advise clients to create stronger links with the object they sponsor as a means to increase brand awareness and/or create better brand image and attitudes (Johar et al. 2006; Verity 2002). The most common types of linkage-creating activities include the placement of the sponsor logo on the team uniform or the use of hospitality areas during the event to host sponsor clients. While these types of activities typically increase the sponsorship rights fees substantially, no empirical research exists to answer the question about the degree to which these linking tactics influence the relative effectiveness of the two mediums.

RQ4: Do activities that link the sponsor more closely with the object influence the relative effectiveness of sponsoring versus TV advertising?

Method

To ensure that each medium would be treated in a fair manner while providing a realistic testing environment, a panel of leading media agencies and broadcasters, media consultants, and market researchers specializing in sponsoring and advertising were consulted extensively in all phases of designing and implementing the experiment.

Stimuli development

The stimuli developed were nine short films produced specifically for the current research using professional team handball as the context, which was the 2nd most popular team sport in Norway where the data was collected. The films simulated the game highlight segments frequently shown during TV and Internet sports news programs. The commercial television station with broadcast rights to professional handball league games supplied the camera crew, equipment, and play-by-play broadcast team to provide the visual and audio inputs for the game coverage portion of the films, while perimeter boards for the filmed game were supplied by the league's official printing firm.

Only two test brands were given sponsorship exposure during each film with each occupying the two highly visible court corners where the TV camera spent most of its

time to capture game action around the goal areas. In addition to the corners, each of the test brands also occupied two other perimeter board spots, with the remaining 12 spots equally allocated to 3 filler brands not part of the study. Figure 1 (top) displays a still shot from one of the films that shows the type of exposure that the test brands received on the perimeter boards.

To provide clean sponsoring to TV ad comparisons, all of the selected test brands needed to meet the following criteria: 1) they were not current or recent sponsors of the teams, 2) they were not direct competitors with each other or with any of the "filler" brands used on the court perimeter, player uniforms, or TV advertising, and 3) they could provide a high quality 30-second TV spot to be used in the films. After discussion with sponsorship consultants that had knowledge of the target audience characteristics of team handball in Norway, the following brands were

Perimeter Board Examples from Film Stimuli



Interview with Uniform Logo from Film Stimuli



Figure 1 Perimeter board examples from film stimuli. Interview with uniform logo from film stimuli.

selected as high fit sponsors: 1) G-Sport: the largest Scandinavian sporting goods chain, and 2) Adidas. Test brands selected as the known, but lower fitting sponsors were: 1) Solo: the dominant Scandinavian fruit flavored soft drink brand, and 2) Coca-Cola which does not emphasize its Olympic sponsorships in the Scandinavian markets. Test brands for the unknown sponsors were 1) AlmBrand, and 2) Codan, which were both Danish insurance companies with no operations in Norway and hence largely unknown to the Norwegian sample (fit was not important for the unknown brands because the experimental subjects would have no basis for evaluating fit). Pre-testing was done on the selected brands as handball sponsors to ensure that they were sufficiently different on perceived fit, and the results showed the lowest high fit brand mean was 5.46, while highest mean for a low fit brand was 2.32 ($p < .001$). RQ4 was tested by the placement of the sponsor logo on the head coach's warm-up uniform that he wore only during a post-game interview conducted by the professional TV announcer (see Fig. 1 bottom). The three scripted interviews (one for each test brand) provided tighter control of the exposure levels and eliminated continuity problems for film editing purposes, while also providing equal jersey location prominence for each of the test brands.

To estimate the relative effectiveness of sponsoring exposure in comparison to TV advertising, multiple levels of sponsoring exposure were needed to determine a response function. In consultation with our panel of experts, three levels of exposure were deemed realistic in representing the typical exposure times that dominant sponsors receive versus less dominant sponsors while also allowing enough variance to calculate a conversion ratio. Therefore, three films were edited for each test brand by a firm specializing in the preparation of television programming, and brand exposure on each film averaged 14%, 25%, and 51% of the game coverage minutes. Two segments of the actual game did not use any of the test brands, but were entirely made up of five filler brands each shown on four perimeter boards in same pattern used for the test brand game segments. Shorter and longer sections of footage from the filler segments of the game were inserted throughout the game coverage during the editing process to create the various exposure levels for each test brand in the finished films, which further created a more realistic environment as perimeter board spots are frequently changed throughout games to provide exposure time for additional sponsors.

Each of the films was 10 min long and included 6 min and 30 s of highlights from a team handball game involving two Norwegian elite division teams plus advertisements in the following order: part 1) 30-second TV spots for three filler brands, part 2) three 5 s TV billboards (i.e. "this program sponsored by ____" to provide realism, but not for

comparison calculations), part 3) the game coverage with perimeter board exposure, part 4) the coach's interview with uniform logo exposure, part 5) a repeat of the same three TV billboards, and 6) three additional 30-second TV spots with the last two being for the test brands. Four test brands were used in each film, but each on only one communication channel (30-second spot OR sponsoring). This meant that all six brands appeared as both TV advertisers and sponsors, but not on the same film, which allowed each brand to be compared with itself across the two mediums (i.e. the results of Coke the advertiser are compared with Coke the sponsor).

To reduce the quality variability of the 30-second spots, the advertising agencies representing the chosen test brands were asked to provide an example of the most effective ad they had recently used in a campaign, but not one that was currently in use in the Norwegian market. Both ads for the unknown brands used the Danish language, but the written and spoken words were the same as Norwegian since the two languages are virtually identical in vocabulary and grammar.

Subjects

Respondents were 2333 Norwegian citizens representing the overall target market populations of the sponsors in the Norway, and recruited from a large consumer panel whose members had agreed to participate in a certain number of Internet based surveys and advertising tests per year in exchange for prize drawing eligibility. While the representativeness of Internet based surveys has been questioned, the panel from which the sample was drawn was specifically recruited for purposes of providing representative samples of common Norwegian target markets by the largest ad and sponsor testing agency in Scandinavia, which is possible to achieve because of the 70% home broadband penetration rate in Norway (Teleplan 2008). Respondents who reported problems with the video/audio and/or prior knowledge about the unknown brands were dropped from the subsequent analysis, resulting in a final sample size of 1,818 and a response rate of 78%.

Experiment procedure

As a cover story, respondents were told that the study was about team handball television coverage. After being randomly shown one of the nine films, respondents were given a questionnaire. The first questions were comprised of advertiser/sponsor brand recognition measures (1 in 4 multiple choice as employed by Johar and Pham 1999 with all four brands being well-known in the known sponsor product category, and unknown brands for the unknown sponsors), and recognition certainty (0–total guess $\leftarrow \rightarrow$

100% certainty). This was followed by questions regarding advertising/sponsoring induced changes to brand liking and purchase intentions using three 7-point scale items adopted from Close et al. 2006; Miniard et al. 2006, with purchase intentions being calculated as the percentage of respondents indicating a 4 or higher response to “likelihood of doing more business with sponsor/advertiser” (Cronbach Alpha’s for the liking scale was .88 or above). Respondents were not allowed go back and change their answers to previous questions or see any part of the film a second time. Since sponsorship effects such as liking and purchase intentions are not likely to be realized if audience members are not able to accurately identify the sponsor (or advertiser) (Johar et al. 2006), only subjects who correctly identified the sponsor and advertiser were used for calculating the liking and purchase intention scores for each test cell (cells ranged from 66% to 84% recognition accuracy). Questions related to each of the four test brands (two advertisers and two sponsors) were randomly rotated to prevent order effects as an alternative explanation. Analysis revealed no significant order effects, and as a result all subsequent analysis used combined groups.

At the end of the questionnaire, respondents were asked about: 1) their interest in handball using a 2-item involvement scale adopted from Mittal and Lee (1989), 2) their ability to clearly see the films on their computer systems, and 3) whether they had prior knowledge of the unknown brand. Previous sponsorship research has found that higher object involvement could influence the ability to accurately recognize the sponsor, and hence could be a potential alternative explanation for the results (Harvey 2001; Johar et al. 2006). One-way ANOVA with Scheffe post hoc tests, however, revealed no significant differences between any of the comparison groups on involvement ($p < .01$).

Study 1 results

Table 1 provides the base rates for comparing the TV spots versus sponsoring on the four effect measures from which the analyses presented in Table 2 are based (the means of all 3 exposure levels are used in sections 1 to 4). Table 1 section 5 presents the results from sponsoring relative to TV ads for each of the four effect measures, and with the exception of Brand Liking, finds a gradual increase in sponsoring effects at increasing exposure levels.

Table 2 section 1 presents the linear regression results used to find the response function necessary for the calculation of the intercept point when sponsoring equals the effects attained by advertising the same brand with a TV spot. Using four predictor variables, the adjusted explained variance ranges from 87.4% to 97.2%. Table 2 section 2 uses the regression coefficients to calculate the number of

sponsor exposure seconds required to equal 30 s of television advertising under the various conditions (i.e. known or unknown sponsor/advertiser, etc.), and the corresponding conversion figure (i.e. the 24.6% average conversion factor for all brands on the recognition measure is calculated by dividing 30 s of TV ad exposure by the 122 s of sponsoring exposure required to achieve the same recognition level as a single 30 s TV spot, which means that it would require almost 4 s of sponsor exposure to equal the recognition achieved from 1 s of TV advertising).

The results related to RQ1 suggest that sponsoring provides stronger performance relative to TV advertising for liking/purchase intentions versus recognition. According to Table 1 section 5, sponsoring provides a mean Recognition Accuracy of 90.9% relative to TV advertising, while the mean for the Liking and Buying Intention cells is a significantly higher 95.9% ($p < .01$). The regression based conversion figures reported in Table 2 section 2 also similarly favor the Liking and Buying Intention effects, which at 142 s and 134.8% are significantly better than the Recognition Effects average of 166 s and 29.6% ($p < .01$).

The results related to RQ2 show that sponsorships are less effective for unknown brands than for known brands relative to TV advertising. This is seen in Table 2 section 1, where the Unknown Brand predictor has a significant negative relationship to Recognition Accuracy, Certainty, and Buying Intention, indicating unknown brands perform poorly relative to known brands. The relationship with Brand Liking is in the same direction, but is not significant. The results reported in Table 2 section 2 further show that Unknown (and unrelated) brands need 365 s across the four effect measures to equal the results of 30 s TV spots for the same brands, significantly longer than the 117 s needed for the known-unrelated brands ($p < .01$). The corresponding conversion rate for unknown brands is 8.2%, which is significantly lower than the 25.6% for known brands ($p < .01$).

The RQ3 results show that sponsoring is relatively more effective versus advertising for lower fit brands when the communication goal is brand recognition, but the opposite when the communication goal is brand liking and increased purchase intentions. Table 2 section 1 shows that the Fit variable is a significant negative predictor of both the Recognition Accuracy and Recognition Certainty measures, indicating that higher levels of fit create relatively lower recognition and recognition certainty. In Table 2 section 2 the projected TV equivalencies show that related brands need 247 s and 203 s to achieve the recognition accuracy and certainty results that the same brands achieved from a 30 s TV spot. The equivalent figures for the known less related brands were significantly less ($p < .01$) at 123 and 91 s respectively. In contrast, Table 2 section 1 shows the Fit variable is a significant positive predictor of both Brand

Liking and Purchase Intentions. Section 2's TV equivalencies further illustrate the positive impact of fit, as related brands needed 2 and 37 s to achieve the Liking and Purchase Intention levels achieved with the same brands using 30 s TV spots, while lower fitting brands needed a significantly higher 113 s on Brand Liking and 140 s on Purchase Intentions ($p < .01$). These results suggest that high fit may reduce encoding of the sponsor in terms of recognition (Jagre et al. 2001), but lead to easier acceptance of the positive sponsor message among those that do encode it (Rifon et al. 2004).

The RQ4 results indicate that the placement of the sponsor logo on the team uniform provides better results, as the Uniform variable is a significant and positive predictor of all effect measures except Brand Liking, where the relationship was positive, but not significant (see Table 2 section 1). Using these results to calculate the TV

advertising equivalencies for the same brands, shows that sponsors with both perimeter board and uniform exposure need an average of 93 s across the four effect measures to achieve the same results as 30 s of TV advertising, while the equivalent figure for brands with only perimeter board exposure is a significantly higher 158 s ($p < .01$). This calculated time difference is 45 s greater than the actual time difference in logo exposure caused by the coach's interview, (which added 20 s exposure for the Uniform brands).

Study 1 conclusion

The sponsoring to TV advertising conversion rates ranged between 5.3% for unknown sponsors on Buying Intentions to 1500% for high fit sponsors on Brand Liking. Although the vast majority of conversion rates fall within the 5% to

Table 1 Sponsor perimeter board results versus TV Ads*

Section 1: All Six Brands		Advertiser		Sponsor	
Recognition Accuracy %	80.89% 2		73.67%		
Certainty of Recognition	7.33	2	6.59		
Sponsor Brand Liking	2.55		2.55		
Positive Purchase Intention %	32.60%		29.93%		
		Unknown Brands		Known Brands	
Section 2:	Advertiser		Sponsor		
Recognition Accuracy %	74.23%		66.12%		
Certainty of Recognition	6.92	2	5.38		
Sponsor Brand Liking	2.36	2	1.98		
Positive Purchase Intention %	28.96% 2		13.72%		
Section 3:	Known Related Brands		Known Low-Related Brands		
Recognition Accuracy %	84.32% 2		71.01%		
Certainty of Recognition	7.65	2	6.70		
Sponsor Brand Liking	3.20	2	3.50		
Positive Purchase Intention %	47.80%		52.17%		
Section 4:	Uniform Brands		No-Uniform Brands		
Recognition Accuracy %	79.28%		78.18%		
Certainty of Recognition	7.25		7.23		
Sponsor Brand Liking	2.57		2.66		
Positive Purchase Intention %	34.90%		32.85%		
Section 5:	All 6 Brands Sponsor % of TV Spot				
Exposure Condition:	Low		Medium		High
n	627		591		600
Seconds of Exposure	56		98		200
Recognition Accuracy %	82.8% 3		88.3% 3		104.0%
Certainty of Recognition	78.1% 2,3		93.7%		98.4%
Sponsor Brand Liking	100.4%		99.5%		99.2%
Positive Purchase Intention %	84.7% 2,3		96.3%		95.5%

*Same brands used in all advertiser versus sponsor comparison groups; 2 or 4 indicates that mean is significantly different than mean in column 2 or 4 at $p < .05$

Table 2 Study 1 sponsorship to TV Ad conversion calculations

Section 1:	Linear Regression Results (2)			
	Recognition		Effect on Brand	
Dependent Variable:	Accuracy	Certainty	Liking	Intention
Predictors Coefficients: (1)				
Constant	71.77	5.99	1.15	15.56
Exposure Time (sec) <i>B</i>	.13*	.01*	.002*	.039*
Unknown Brand (dummy) <i>B</i>	-30.37*	-3.15*	-.033	-8.55*
Fit of Brand (dummy) <i>B</i>	-17.18 *	-.77	.972*	30.78*
Uniform (dummy) <i>B</i>	18.96*	2.18*	.039	4.16*
Adj. Explained Variance	87.4%	89.1%	97.2%	93.8%

(1) non-standardized regression coefficients; (2) minimum tolerance level for any of the models was .499, while the maximum VIF was 2.00;
 * predictor is significant at $p < .05$ (1-tailed)

Section 2:	Projected Sponsor Exposure Seconds to Achieve 30-second TV Ad Results				
	Recognition		Effect on Brand:		
	Accuracy	Certainty	Liking	Intention	Avg.
All Brands	122	128	93	165	127
Unknown Brands	327	340	230	563	365
Known Brands (50% related)	91	161	60	89	100
Unrelated Brands (known brands)	96	120	113	140	117
Related Brands (known brands)	234	202	2	37	119
Brands not on Uniform	208	226	113	83	158
Brands on Uniform & Perimeter Board	34	31	104	201	93
Category Average	166		142		

	Per Second Value of Sponsorship Relative to Per Second TV Ad Results				
All Brands	24.6%	23.4%	32.3%	18.2%	23.6%
Unknown Brands	9.2%	8.8%	13.0%	5.3%	8.2%
Known Brands (50% related)	33.0%	18.6%	50.0%	33.7%	30.0%
Unrelated Brands (known brands)	31.3%	25.0%	26.5%	21.4%	25.6%
Related Brands (known brands)	12.8%	14.9%	1500.0%	81.1%	25.2%
Brands not on Uniform	14.4%	13.3%	26.5%	36.1%	19.0%
Brands on Uniform & Perimeter Board	88.2%	96.8%	28.8%	14.9%	32.2%
Category Average	29.6%		134.8%		

Note: all brands displayed on perimeter boards only unless described explicitly as also on uniform.

100% “rule of thumb” levels used in the “black-boxes” of many practitioners (Cornwell et al. 2005; Crompton 2004), the wide variance does not support the common industry practice of using a single “one-size-fit-all” conversion rate. The results show that known brands benefit more than unknown brands from sponsoring relative to TV advertising. The situation is a bit more complicated for high fit sponsors versus lower fit sponsors, however, as related brands are disadvantaged on measures of Sponsor Recognition, but have a significant advantage on Brand Liking and Purchase Intentions. Furthermore, logo placement on the team uniform provided significant advantages on all effect measures, except Brand Liking, that was over and

above the added audience exposure time to the brand. Overall, the results suggest that the financial value of sponsoring varies greatly depending on the type of brand/product category and/or the goals of the sponsor.

Study 2: Sponsorship and TV advertising synergy

Another untested but common industry “rule of thumb” suggests that sponsoring need to be “activated” with an advertising budget that is double or triple the amount spent on sponsorship rights (Cornwell et al. 2005; Verity 2002). The advertising is thought to enhance the effec-

Table 3 Study 2 sponsorship and TV Ad synergy effects

		3 Brands			
Section 1:	Sponsor Only	Sponsor + TV ad			
n	411	411	t or z / p		
Seconds Exposure	75	105	1.743 / .042		
Recognition Accuracy %	84.8%	108.6%	10.05 / .000		
Certainty of Recognition	84.4%	99.3%	3.42 / .000		
Sponsor Brand Liking	107.1%	107.1%	0.00 / .999		
Positive Purchase Intention %	96.3%	121.9%	9.16 / .000		
Section 2:	Linear Regression Results (2)				
	Recognition		Effect on Brand:		
Dependent Variable:	Accuracy	Certainty	Liking	Intention	
Predictors Coefficients: (1)					
Constant	67.76	6.84	1.46	21.84	
Exposure Time (sec) <i>B</i>	.085*	.007*	.000	.029	
Unknown Brand (dummy) <i>B</i>	-20.153*	-3.270*	-.157	-7.218	
Related Brand (dummy) <i>B</i>	- 2.244	-.656	.873*	24.539*	
TV&Sponsorship (dummy) <i>B</i>	12.550*	.608*	.608*	2.388	
Adj. Explained Variance	77.3%	91.5%	85.4%	69.3%	
(1) non-standardized regression coefficients; (2) minimum tolerance level for any of the models tested was .559 and maximum VIF was 1.78; * predictor is significant at <i>p</i> <.05 (1-tailed)					
Section 3:	Projected Sponsor Exposure Seconds to Achieve 30-second TV Ad + 75 seconds of Sponsor Exposure Results				
	Recognition		Effect on Brand:		
	Accuracy	Certainty	Liking	Intention	Avg.
All Brands	251	191	na	187	183
Unknown Brands	263	216	na	504	312
Unrelated Brands (known brands)	220	153	na	86	170
Related Brands (known brands)	260	211	na	1	183
	Per Second Effects of Sponsorship Alone Relative to Per Second Synergy Effects				
All Brands	42%	55%	na	56%	57.4%
Unknown Brands	40%	49%	na	21%	33.7%
Unrelated Brands (known brands)	48%	69%	na	122%	88.2%
Related Brands (known brands)	40%	50%	na	1500%	82.0%

tiveness of sponsoring for a variety of reasons including: 1) increased exposure intensity (Lardinoit and Derbaix 2001; Olson and Thjømøe 2003), 2) increased ability to explain the fit between sponsor and object in cases where there is not a “natural fit” (Cornwell et al 2006; Simmons and Becker-Olsen 2006), and 3) to prevent guerrilla marketers from buying advertising during the sponsored event and thereby stealing “recognition” as the official sponsor (Crimmins and Horn 1996, Meenaghan 1996; Turner 2004). None of this literature, however, has answered the research question regarding the level of synergistic

communication effects derived by activating sponsorships with TV advertising:

RQ5: Does the combination of sponsoring with TV advertising create synergy between mediums versus sponsoring alone?

Method

Study 2 used virtually the same stimuli and measures as study 1. Due to budgetary reasons only 3 brands were used

in study 2: AlmBrand, Coca-Cola, and Adidas. Furthermore, due to likely ceiling effects, only the two lowest sponsor exposure levels were tested (approximately 14% and 25% of the game coverage minutes). Using the same handball game footage and TV spots as in study 1, the study 2 films were identical with the exception that one of the test brands was used twice in each film as both a sponsor and TV advertiser. One limitation, which might decrease the potential synergy effects, was that none of the 30-second spots discussed or promoted the handball sponsorship due to the prohibitive costs of creating professional looking ads for the experimental stimuli.

Subjects were 1025 Norwegian citizens randomly assigned to see one of the six films, and recruited from the same consumer panel as study 1, although no subjects from study 1 were also used in study 2. Subjects were excluded if they indicated problems seeing the video and/or had prior knowledge about the unknown brand, which resulted in a final total of 822 subjects available for data analysis (80% response rate). No significant differences were found between the groups on object involvement, nor were any significant order of presentation effects found.

Study 2 results

Table 3 section 1 compares the sponsorship only results with the sponsorship + TV ad results, and finds significant improvements with the addition of a 30-second spot on all but the Brand Liking measure. A TV&Sponsorship dummy variable was added to study 1's linear regression model to determine if this improvement was from synergy or just the added exposure time (the dummy variable Uniform was also dropped due to multicollinearity problems). Table 3 section 2 shows the results for RQ5 as the TV&Sponsorship variable is a significant positive predictor of three measures of communication effects, indicating that the two mediums together provide some improvements that are not accounted for by the longer exposure time to the brand. Exposure Time in this setting is not a significant predictor for either Brand Liking or Purchase Intentions, and nominally has no impact at all on Brand Liking.

Table 3 section 3 uses the regression results to answer RQ5 by projecting the number of seconds of sponsoring exposure required to equal the results of one 30-second TV spot + 75 s of sponsor exposure (i.e. 105 s of total brand exposure). Since Exposure Time was not even a nominal influence on Brand Liking, no equivalency values for this effect could be calculated. An average of 183 s of sponsor exposure was required to equal the effects of 105 s of sponsor and TV ad exposure across the three stimuli brands and remaining three effects, which implies a conversion rate of 57.4% and supports some synergy effect overall. Within this 183 s average is a great deal of variance, however,

ranging from 1 s on purchase intention for related brands, to 504 s on purchase intentions for unknown brands.

Study 2 conclusion

The study 2 results indicate there is a significant synergy effect from combining TV advertising with moderate amounts of sponsoring exposure on three out of four communication effects. Unknown brands benefit most with the addition of advertising, since they need the highest level of sponsor exposure to equal the “synergy” condition (312 s on average). The financial implications of this will depend on the pricing structures for advertising and moderate versus heavier sponsorship coverage, because if heavy sponsoring exposure can be attained at less than 57.4% of the cost of the synergy conditions, using sponsoring alone will offer the better return, although this is also subject to variations depending on type of goal and sponsor. The results, however, do not support the general industry rule of thumb that the cost of the sponsoring rights should be multiplied by 2 or 3 times to “activate” the sponsorship with brand advertising, unless perhaps the dangers from guerilla marketing activities is high.

Discussion

Although managers are often faced with multi-million dollar decisions regarding sponsorships, little is known about the veracity of TV advertising conversion rates used inside the “black boxes” and other sponsoring industry rules of thumb. By using realistic sponsorship stimuli, target market representative samples, and four measures of communication effects, study 1 found that it took 127 s of logo exposure during a sponsored game to equal the effects created by advertising the same brands using a standard 30-second spot, implying a conversion rate of 24%. This overall average conversion rate across all the effect measures, however, disguises some rather large variations within the sample such as between unknown and known sponsors at 8.2% versus 30.0% respectively. The same type of sponsor could also face huge differences in relative effectiveness depending on the communication goal, with high fit brands having a 12.8% conversion rate on Brand Recognition, but 1500% on Brand Liking. These differences have potentially huge financial implications. For example, the 93 s average conversion rate for the uniform brands versus 158 s for non-uniform brands suggests that logo placement on the uniform in addition to perimeter boards is worth about 41% extra money $(1-(93/158)=.411)$, but this bonus factor would be 83% for recognition goals and minus 42% for purchase intention goals.

How should these results be used to calculate the financial value of sponsoring? The simple answer is each

30-second spot equivalent of sponsoring exposure should be valued at the financial cost of a 30-second spot to reach the same audience (i.e. running the ad during the sponsored event). As an example, assume that a known lower-fit sponsor has a communication goal of increasing purchase intentions. The appropriate sponsoring conversion rate from Table 2 section 3 is 21.4%. Taking this figure and assuming 500 s sponsor exposure during a 3-hour game would be the equivalent of 3.6 TV 30 s spots (i.e. $(500 * 21.4\%) / 30 = 3.57$). Using typical Norwegian market figures, the ad spot cost to reach a handball TV audience of 200,000 people is priced at about \$8,000 (per spot), while the production of a 30 s spot might cost \$100,000. We will assume this spot will run during 20 games, making the ad spot production cost \$5,000 per game (i.e. $\$100,000 / 20$). The sponsorship rights cost per handball game for 4 perimeter boards would be about \$7,500, while the perimeter board production cost would be about \$200 (i.e. the costs to produce the 4 boards). Plugging these numbers into the sponsorship relative value formula: $(1 - (((3.6 * \$7,500) + \$200) / ((3.6 * \$8,000) + \$5,000)))$ results in a value of 19.6% and suggests that the sponsorship provides almost 20% better value with regards to purchase intentions than the equivalent amount of exposure using 30 s TV ads. Conversely, a negative value would signal that TV advertising was more efficient than sponsoring at achieving a particular goal for a particular type of sponsor/advertiser, and might further suggest that the sponsorship rights are overpriced.

Study 2 examined claims that using sponsoring together with advertising might create some synergistic benefits. For three of the four communication effect measures, the synergy variable was a significant positive predictor over and above the impact of higher exposure levels that came with adding the 30-second spot to the sponsor exposure. Across the three communication effects, it took 183 s of sponsor exposure to equal the effects created by one 30-second spot together with 75 s of sponsor exposure, implying a conversion rate of 57% (i.e. $(75+30) / 183 = .57$). Once again, the variability was high around the mean, with a conversion rate of 34% for unknown brands and 88% for known-unrelated brands. The most extreme case was for related-known brands with the goal to increase buying intentions, with a conversion rate of 1500%, indicating the addition of the advertisement actually lowered the desirability of the brand.

Limitations

The sponsoring to TV advertising conversion rates generated in the two studies relied on 10 min “game highlight” films that gave respondents only one exposure to each test brand TV spot. The suggested sponsoring valuation calculations assume the conversion rates from the short duration stimuli would remain the same when exposure levels

to both perimeter boards and TV spots might be considerably higher during a full game. This assumption is based on previous research that finds exposure/advertising effects are not linear, but flatten out after a few exposures (Bornstein 1989; Jones 2006; Vakratsas and Ambler 1999). Use of the conversion rates found here would therefore be fairly accurate if both mediums would be similarly effected by flattening response curves over “full-game” time periods, although future research might use longer exposure periods to test this assertion.

A second limitation of the methodology is its probable favoring of advertising. While panel respondents in the current studies had a lottery draw incentive to view the stimuli TV ads, previous research has established that TV viewers often ignore/eliminate advertisements by channel surfing or leaving the room (Crompton 2004; Harvey 2001; Verity 2002). Furthermore, the test brand TV ads were selected by advertising agencies because they were particularly effective and they were also the last things that subjects saw on the film before answering the questionnaire (which is also typical of game coverage where the ads come after the game). This means subject exposure was freshest for the good quality TV spots, which might slightly enhance advertiser recognition and certainty. Another element that probably understates the effects of sponsoring is the exclusion of its hospitality component, which is a growing area of emphasis in building relationships between the sponsor and important clients (Verity 2002). Altogether, these factors likely mean that the conversion rates are conservative since the demonstrated effects of the stimuli TV ads are probably higher than the general population of TV ads would achieve in the real world, while the sponsoring effects may be understated. Future research might examine how the conversion rates are affected when the TV ads are of variable quality and not the last items seen by subjects.

A third limitation is the use of a Scandinavian sample and a single sports context. The fact that the calculated conversion rates found were mostly within the ranges previously advocated, but not empirically tested, suggests that this limitation may not be serious. Certain contexts, however, might create further deviations than were found in the current studies. For example, how might the value of logo placement on player uniforms vary in slow moving sports such as golf or fast moving sports such as auto racing? Or how might less intensive perimeter boards placed further away from the “playing field”, such as often seen in golf tournaments and some non-sports sponsorships, impact conversion rates? In addition, only extremely well known or totally unknown brands were used as stimuli, and it might be possible that the relative effectiveness of sponsoring is not linear between these extremes. Future research might try to replicate the current studies to determine how the relative effectiveness of sponsoring varies across markets, brand strengths, and other sports and non-sports contexts.

Finally, study 2 is probably conservative in assessing the synergy value of using the two mediums together. Since the study design made it virtually impossible to use “real” advertisements that promoted the fictional sponsorships used as stimuli, it is very likely that the synergy effects would be higher if the ads could be used to explain the sponsorship or tie in with the sponsorship more directly by using the sponsored athlete as an advertising spokesperson. This would be particularly true for sponsoring brands that are unknown and/or perceived as “naturally” low in fit, as previous research has found that explanations of fit do improve sponsoring results in such situations (Cornwell et al 2006; Simmons and Becker-Olsen 2006). Future research might attempt to find real sponsorship contexts with supporting TV spots to determine the synergy effect when ads specifically promote the sponsorship, but the added costs of creating sponsorship specific TV spots would also need to be included in the synergy valuations.

Conclusion

Despite the limitations of the two studies, they are the most comprehensive and realistic empirical comparison between sponsoring and TV advertising ever attempted. Although the de facto standard for valuating sponsor exposure is to convert them into TV advertising equivalencies, no previous study has attempted to verify the inner workings of the black-boxes used for such purposes. In general, the results clearly show that relatively short exposures to perimeter boards can provide comparable recognition, liking, and intention effects of TV advertising. Contrary to industry practice, the variations in TV ad equivalencies found in the studies indicate that black-boxes will need to use multiple conversion rates to provide accurate sponsoring valuations that account for variables such as sponsor communication goal, fit between sponsor and object, prior brand awareness, and level of sponsor visibility/connection.

By thoughtful use of the conversion rates from these studies, managers involved with advertising or sponsoring should be more confident in calculating which medium or combination of mediums offers the best value, and as a consequence have a greatly improved basis for making communication budget allocations and negotiating contracts with purveyors of both mediums.

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